

Advanced Structural Analysis using ANSYS

Lecture and Exercise Notes

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Exercise 3: Elasto-plastic deformation

Die Veranstaltung basiert u.a. auf dem Vorlesungs- und Übungsskript „Die Methode der finiten Elemente“, gehalten durch Prof. Dr. rer. nat. M. Braun, Dr.-Ing. H. Walter, Dr.-Ing. U. Zwiers und Dr.-Ing. Philipp Olle, am Lehrstuhl für Mechanik (Maschinenbau) der Universität Duisburg-Essen im WS2004/05

General Remarks

ANSYS files

The command `/FILNAME, Fname, Key` changes the Jobname, here `Fname`, for the analysis. Based on this, several output files will be generated during the run of ANSYS in your work directory. The *work directory* is

- specified during the start of ANSYS by the Launcher or
- changed in the ANSYS GUI by `>File → Change directory<` or
- changed by APDL command `/CWD, 'path'`¹

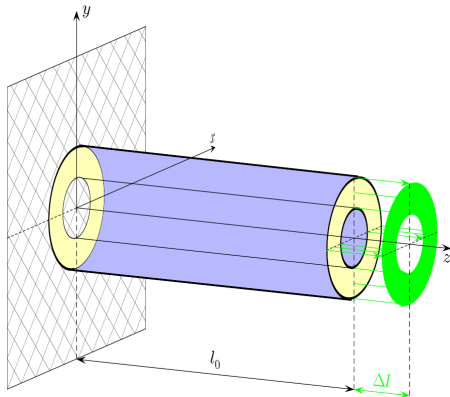
The *output files* generated in the work directory are

- `Fname.db`: Database file (binary). This file stores the geometry, boundary conditions and any solutions.
- `Fname.dbb`: Backup of the database file (binary).
- `Fname.err`: Error file (text). Listing of all error and warning messages.
- `Fname.out`: Output of all ANSYS operations (text). This is what normally scrolls in the output window during an ANSYS session.
- `Fname.log`: Logfile or listing of ANSYS commands (text). Listing of all equivalent ANSYS command line commands used during the current session.
- all other files: Depending on the operations carried out, other files may have been written. These files may contain results, etc.

¹if directory `path` not exist, it must be create by `/SYN, mkdir, 'path'`

Cantilever (elastoplastic material behavior) with end displacement

Problem 1



- Geometric data

$$l_0 = 1000 \text{ mm}$$

$$r_{a0} = 80 \text{ mm}$$

$$r_{i0} = 40 \text{ mm}$$

- Material data (elastic domain)

$$E = 3000 \text{ N/mm}^2$$

$$\nu = 0.2$$

- Clamping

Fixed support at one end of the tube
Deformation of cross section possible

- Load case

Displacement in two load steps (LS)

1st LS: $\Delta l = 50 \text{ mm}$

2nd LS: Remove displacement
condition from 1st LS

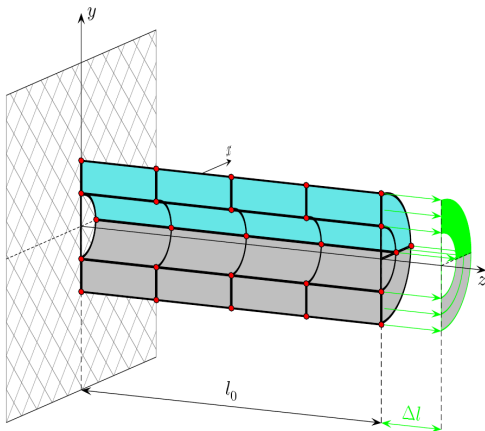
- Objective target

σ_z as function of Δu_z

relaxed geometry l_1, r_{a1}, r_{i1}

3D model, boundary condition, 1st load case

Cantilever with end displacement, Problem 1



• Element types

2-D PLANE182 & 3-D SOLID185

(Simplified enhanced strain formulation)

• Boundary conditions

Nodes	Condition
$x = 0$	$u_x = 0$
$y = 0$	$u_y = 0$
$z = 0$	$u_z = 0$
$z = l$	$u_z = \Delta l$

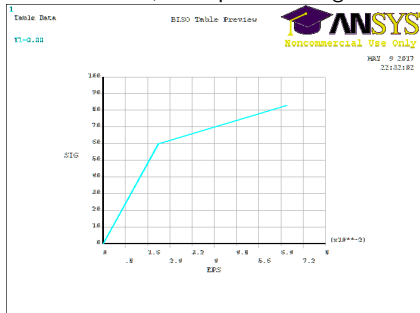
• Modelling and meshing

- Modelling a quarter cross section
- Meshing the quarter area
(4 elements along arc length,
2 elements along tube wall thickness)
- Extrusion to a 3D half model with
20 elements along the tube length
- Removing the elements of the base
area ($x = 0$)
- Apply displacement in 50 steps

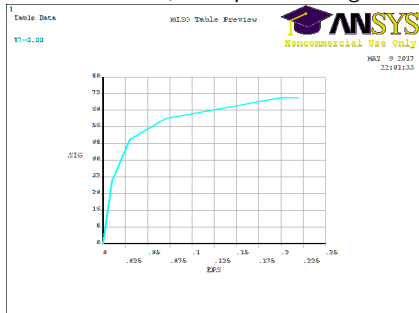
Material behavior: von Mises Elastoplasticity

Cantilever with end displacement, Problem 1

bilinear, isotropic hardening



multilinear, isotropic hardening



elastic domain

Young's modulus E in N/mm^2 3000 Possion's ratio ν 0.2

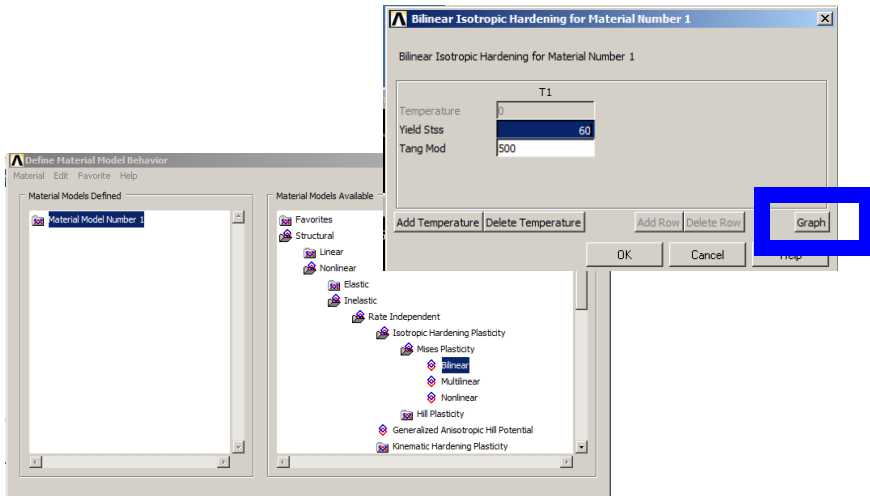
plastic domain

hardening type	bilinear	hardening type	multilinear
yield point	σ_y in N/mm^2 60	base point 1	ε_1, σ_1 in N/mm^2 (0.01, 30)
hardening slope	h_y in N/mm^2 500	base point 2	ε_2, σ_2 in N/mm^2 (0.03, 50)
		base point 3	ε_3, σ_3 in N/mm^2 (0.07, 60)
		base point 4	ε_4, σ_4 in N/mm^2 (0.20, 70)

Material behavior: Graph stress vs. strain

Cantilever with end displacement, Problem 1

Main Menu → Preprocessor → Material Props → Material Models



Additional ANSYS APDL Commands: Pre Processor

Cantilever with end displacement, Problem 1

Pre Processing /PREP7

material law (elastic domain) MP, . . .

material law (plastic domain): bilinear, isotropic hardening

select material model TB, . . .

input parameter TBDATA, . . .

material law (plastic domain): multilinear, isotropic hardening

select material model TB, . . .

input parameter TBPT, . . . **Multilinear hardening needs plastic strains as input parameters**

more help: help,XXX, whereby XXX should be replace by command.

Additional ANSYS APDL Commands: Solution Processor

Cantilever with end displacement, Problem 1

Solution Processing

/SOLU

entering 1st LS

TIME,1 (as this bvp is time independent, you can use any positive value for time)

...

e.g. D, . . . set up boundary condition

e.g. F, . . . load

e.g. NSUBST, . . . and AUTOTS, . . . substep parameter (50 substeps !)

e.g. OUTRES, . . . to control solution data written to database

...

solving 1st LS

SOLVE

entering 2st LS

TIME,2 (as this bvp is time independent, you can use any positive value greater than the one used in the first loadstep for time)

...

set up boundary condition (delete from 1st LS), load, substep parameter, output control

...

solving 2st LS

SOLVE

more help: help,XXX, whereby XXX should be replace by command.

Additional ANSYS APDL Commands: Time History Post Processor

Cantilever with end displacement, Problem 1

<u>Time History Post Processing</u>	<code>/POST26</code>
define number of variables	<code>NUMVAR,...</code>
define nodal solution data (only the degree-of-freedom results)	<code>NSOL,...</code>
define element solution data (derived element results)	<code>ESOL,...</code>
set variable of x -axis	<code>XVAR,...</code>
set plotting variables	<code>PLVAR,...</code>

General Plot Options

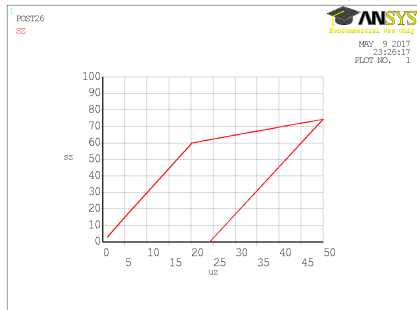
type of grid	<code>/GRID,...</code>
various line options	<code>/GROPT,...</code>
line thickness	<code>/GTHK,...</code>
axes label	<code>/AXLAB,...</code>
x axis range	<code>/XRANGE,...</code>
y axis range	<code>/YRANGE,...</code>
color of various items	<code>/COLOR,...</code>

more help: `help,XXX`, whereby XXX should be replaced by command.

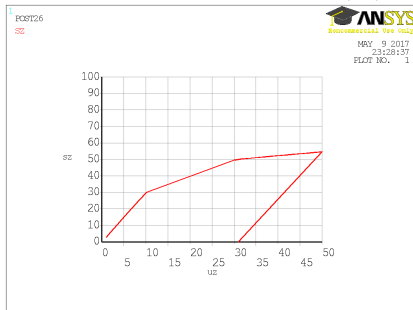
Results

Cantilever with end displacement, Problem 1

a) bilinear, isotropic hardening
stress vs. displacement $\sigma_z = \sigma_z(u_z)$



b) multilinear, isotropic hardening
stress vs. displacement $\sigma_z = \sigma_z(u_z)$

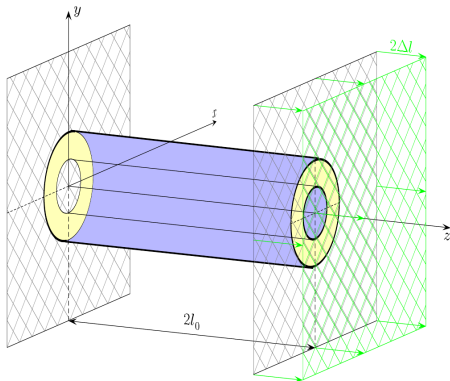


deformations/length after unloading (taken at $z = l_1$, $y = 0$ mm and $x = 40$ mm or $x = 80$ mm)

a)	change of tube length	Δl_1 in mm	24.281	b)	change of tube length	Δl_1 in mm	31.682
	change of outer radius	Δr_a in mm	0.9539		change of outer radius	Δr_a in mm	1.2380
	change of inner radius	Δr_i in mm	0.4770		change of inner radius	Δr_i in mm	0.61898

Cantilever with end displacement

Problem 2



- Geometric data

$$l_0 = 1000 \text{ mm}$$

$$r_{a0} = 80 \text{ mm}$$

$$r_{i0} = 40 \text{ mm}$$

- Material data (elastic domain)

$$E = 3000 \text{ N/mm}^2$$

$$\nu = 0.2$$

- Clamping

Fixed support at one end of the tube
Deform. of cross section *NOT* possible

- Load case

Displacement in two load steps (LS)

1st LS: $\Delta l = 50 \text{ mm}$

2nd LS: Remove displacement
condition from 1st LS

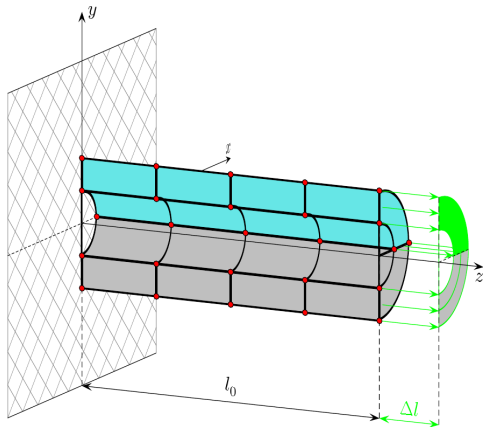
- Objective target

σ_z as function of Δu_z

relaxed geometry l_1, r_{a1}, r_{i1}

3D model, boundary condition, 1st load case

Cantilever with end displacement, Problem 2



• Element types

2-D PLANE182 & 3-D SOLID185
(Simplified enhanced strain formulation)

• Boundary conditions

Nodes	Condition
$x = 0$	$u_x = 0$
$y = 0$	$u_y = 0$
$z = 0$	$u_x = 0, u_y = 0, u_z = 0$
$z = l$	$u_z = \Delta l$

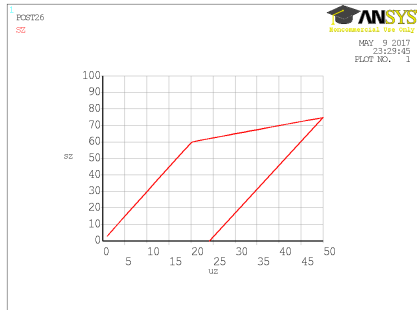
• Modelling and meshing

- Modelling a quarter cross section
- Meshing the quarter area
(4 elements along arc length,
2 elements along tube wall thickness)
- Extrusion to a 3D half model with
20 elements along the tube length
- Removing the elements of the base
area ($x = 0$)
- Apply displacement in 50 steps

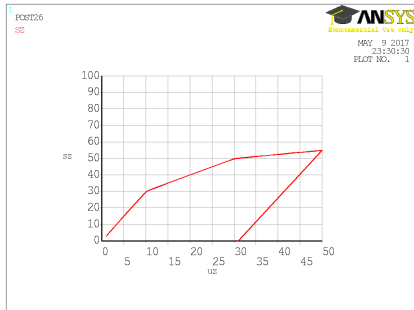
Results

Cantilever with end displacement, Problem 2

a) bilinear, isotropic hardening
stress vs. displacement $\sigma_z = \sigma_z(u_z)$



b) multilinear, isotropic hardening
stress vs. displacement $\sigma_z = \sigma_z(u_z)$



deformations/length after unloading (taken at $z = l_1$, $y = 0$ mm and $x = 40$ mm or $x = 80$ mm)

a) change of tube length Δl_1 in mm 24.219
change of outer radius Δr_a in mm 0.98128
change of inner radius Δr_i in mm 0.4909

b) change of tube length Δl_1 in mm 31.669
change of outer radius Δr_a in mm 1.2719
change of inner radius Δr_i in mm 0.63593